Numerical Analysis: Midterm (**50 marks**)

Solutions

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Question 1 (Floating point arithmetic, **10 marks**)**.** True or false?

1. Let $(\bullet)_2$ denote binary representation. Then $(0.1101)_2 + (0.0011)_2 = (1.0)_2$.
2. Let ε_{64} denote the machine epsilon for the `Float64` format, i.e. `eps(Float64)`. Then the number 2 is representable exactly in this format, and the next representable number is $2 + \varepsilon_{64}$.
3. It holds that $(10000)_2 \times (0.1010)_2 = (1010)_2$.
4. In Julia, `Float64(.6) == Float32(.6)` evaluates to `true`.
5. The spacing (in absolute value) between successive single-precision (`Float32`) floating point numbers is constant.
6. Infinitely many distinct real numbers can be represented exactly in the `Float64` format, but only finitely many can be represented exactly in the `Float32` format.
7. It holds that $(0.\overline{101})_2 = \frac{5}{7}$.
8. Machine addition $\widehat{+}$ is an operation that is *associative* but not *commutative*.
9. The machine epsilon is the smallest number of the form 2^{-n} with $n \in \mathbf{N}$ that can be represented exactly in a floating point format.
10. In Julia, the expression `1 + eps()/3 == 1 + eps()` evaluates to `true`.
11. **Bonus.** In Julia, the expression `exp(eps()/2) == 1 + eps()` evaluates to `true`.
12. **Bonus.** In Julia, the expression `cos(eps()) == 1` evaluates to `true`.

Solution. Below, $\varepsilon = \varepsilon_{64} = 2^{-52}$.

1. **True**: $(0.1101)_2 + (0.0011)_2 = \frac{13}{16} + \frac{3}{16} = 1$.
2. **False**: the number 2 is representable as $2^1 \times (1.00\ldots0)_2$, but the spacing between successive `Float64` numbers in $[2, 4)$ is $2^1 \times 2^{-52} = 2\varepsilon$, so the next representable number is $2 + 2\varepsilon$.
3. **True**: multiplying by $(10000)_2 = 2^4$ shifts the binary point by four positions to the right, so $(10000)_2 \times (0.1010)_2 = (1010)_2$. In decimal notation, $16 \times 0.625 = 10$.

4. **False**: the number $0.6 = (0.1\overline{0011})_2$ has an infinite binary expansion, so it is rounded to 53 significant bits by `Float64` and to 24 bits by `Float32`. The two results are different real numbers, and the comparison, which converts the `Float32` number to `Float64` exactly, returns `false`.
5. **False**: the spacing is $2^E \times 2^{-23}$ between numbers with exponent E, so it doubles at every power of 2.
6. **False**: both formats contain only finitely many numbers, since they are encoded with finitely many bits (64 and 32, respectively).
7. **True**: if $x = (0.\overline{101})_2$, then $2^3 x = (101.\overline{101})_2 = 5 + x$, so $7x = 5$.
8. **False**: it is the other way around. Machine addition is commutative, since $x \mathbin{\widehat{+}} y = \mathrm{fl}(x+y) = \mathrm{fl}(y+x) = y \mathbin{\widehat{+}} x$, but it is not associative, because intermediate results are rounded. For example, in `Float64`, $(1 \mathbin{\widehat{+}} \varepsilon/2) \mathbin{\widehat{+}} \varepsilon/2 = 1$ whereas $1 \mathbin{\widehat{+}} (\varepsilon/2 \mathbin{\widehat{+}} \varepsilon/2) = 1 + \varepsilon$.
9. **False**: much smaller powers of 2 can be represented exactly, for example 2^{-1022} in `Float64`. The machine epsilon is the distance between 1 and the next representable number.
10. **False**: the exact sum $1 + \varepsilon/3$ is closer to 1 than to $1 + \varepsilon$, so `1 + eps()/3` evaluates to 1, which is different from `1 + eps()`.
11. **True**: by Taylor's theorem, $\exp(\varepsilon/2) = 1 + \varepsilon/2 + \varepsilon^2/8 + \cdots$, which is slightly larger than $1 + \varepsilon/2$, the midpoint between the successive floating point numbers 1 and $1 + \varepsilon$. Rounding to the nearest floating point number therefore gives $1 + \varepsilon$.
12. **True**: $\cos(\varepsilon) = 1 - \varepsilon^2/2 + \cdots$. The floating point number preceding 1 is $1 - \varepsilon/2$, and since $\varepsilon^2/2$ is much smaller than $\varepsilon/4$, the nearest floating point number to $\cos(\varepsilon)$ is 1.

△

Question 2 (Interpolation and approximation, **10 marks**)**.** Are the following statements true or false? Prove or disprove. Recall that $\mathcal{P}_d$ denotes the set of polynomials of degree at most d.

1. Assume that $x_0 < x_1 < x_2 < x_3$ and y_0, y_1, y_2, y_3 are given real numbers. Then there exists a polynomial $p \in \mathcal{P}_3$ such that $p(x_i) = y_i$ for all $i \in \{0, 1, 2, 3\}$.
2. Assume that $x_0 < x_1 < x_2$ and y_0, y_1, y_2 are given real numbers. Then, there can exist *at most one* polynomial $p \in \mathcal{P}_2$ such that $p(x_i) = y_i$ for all $i \in \{0, 1, 2\}$.
3. Let $p \in \mathcal{P}_d$ be a polynomial of degree $d > 0$, and let $q \colon \mathbf{R} \to \mathbf{R}$ be given by $q(x) = p(x+1) - p(x)$. Then it holds that $q \in \mathcal{P}_{d-1}$.
4. For $n \in \mathbf{N}$, let $x_i^n = i/n$ for $i = 0, 1, \ldots, n$. Assume that $u \colon \mathbf{R} \to \mathbf{R}$ is the smooth function given by $u(x) = \sin(3x) + x^3$, and let $p_n \in \mathcal{P}_n$ denote the polynomial interpolation of u at the points $x_0^n, x_1^n, \ldots, x_n^n$. Then it holds that

$$\lim_{n \to \infty} \left(\max_{x \in [0,1]} \left| u(x) - p_n(x) \right| \right) = 0 \,.$$

Solution. 1. **True**: the Lagrange interpolating polynomial

$$p(x) = \sum_{i=0}^{3} y_i \, \ell_i(x), \qquad \ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j},$$

belongs to $\mathcal{P}_3$ and satisfies $p(x_i) = y_i$, because $\ell_i(x_j) = 1$ if $i = j$ and 0 otherwise.

2. **True**: if $p, \widetilde{p} \in \mathcal{P}_2$ both satisfy the interpolation conditions, then $p - \widetilde{p} \in \mathcal{P}_2$ has the three distinct roots x_0, x_1, x_2. A nonzero polynomial of degree at most 2 has at most 2 roots, so $p - \widetilde{p} = 0$.

3. **True**: write $p(x) = \sum_{k=0}^{d} a_k x^k$. By the binomial formula, $(x+1)^k - x^k = \sum_{j=0}^{k-1} \binom{k}{j} x^j$ is a polynomial of degree $k-1$ for $k \geqslant 1$, and it is zero for $k = 0$. Therefore $q(x) = \sum_{k=1}^{d} a_k \big((x+1)^k - x^k\big)$ belongs to $\mathcal{P}_{d-1}$.

4. **True**: by the interpolation error formula, for every $x \in [0,1]$ there exists $\xi \in [0,1]$ such that

$$u(x) - p_n(x) = \frac{u^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i^n).$$

Each factor of the product satisfies $|x - x_i^n| \leqslant 1$, and, for $n \geqslant 3$, the derivatives of x^3 of order $n+1$ vanish, so $|u^{(n+1)}(\xi)| = 3^{n+1} |\sin^{(n+1)}(3\xi)| \leqslant 3^{n+1}$. Consequently,

$$\max_{x \in [0,1]} \big|u(x) - p_n(x)\big| \leqslant \frac{3^{n+1}}{(n+1)!} \xrightarrow[n \to \infty]{} 0.$$

Note that Runge's phenomenon does not occur here, because all the derivatives of u are bounded by 3^{n+1}, up to a constant.

△

Question 3 (Interpolation, open-ended question, **5 marks**)**.** In polynomial interpolation, the error depends both on the function being interpolated and the choice of interpolation nodes. Consider two families of nodes on the interval $[-1, 1]$:

- Equally spaced nodes,
- Chebyshev nodes.

Discuss qualitatively (and illustrate with examples or plots if you wish) how the interpolation error behaves as the degree n increases in each case. Why does one choice of nodes perform better for large n, and what is the mathematical motivation for using Chebyshev nodes? *You may refer to Runge's phenomenon, but go beyond merely stating it.*

Solution. Many answers are possible; a good answer should contain the following elements. For a smooth function u and interpolation nodes $x_0, \ldots, x_n$ in $[-1, 1]$, the interpolation error satisfies

$$u(x) - p_n(x) = \frac{u^{(n+1)}(\xi)}{(n+1)!} \, \omega_{n+1}(x), \qquad \omega_{n+1}(x) = \prod_{i=0}^{n} (x - x_i).$$

The first factor depends only on the function u, while the second depends only on the nodes.

- *Equally spaced nodes.* The function ω_{n+1} is small in the middle of the interval but very large near the endpoints, where its maximum is larger than in the middle by a factor growing exponentially with n. If the derivatives of u grow quickly, as for the Runge function $u(x) = (1 + 25x^2)^{-1}$, whose $(n+1)$-th derivative grows roughly like $(n+1)!\,5^{n+1}$, the error grows with n near the endpoints: the interpolant oscillates with increasing amplitude there (Runge's phenomenon). Equivalently, the Lebesgue constant of equally spaced nodes grows like 2^n, so the interpolation can be much worse than the best polynomial approximation, and it is also very sensitive to rounding errors and noise in the data.
- *Chebyshev nodes* $x_i = \cos\left(\frac{(2i+1)\pi}{2(n+1)}\right)$, $i = 0, \dots, n$, are the roots of the Chebyshev polynomial T_{n+1}. They cluster near the endpoints, precisely where equally spaced nodes perform poorly. For these nodes, $\omega_{n+1} = 2^{-n} T_{n+1}$, so that $\max_{[-1,1]} |\omega_{n+1}| = 2^{-n}$, and this is the smallest possible value over all choices of $n+1$ nodes: Chebyshev nodes minimize the node-dependent factor in the error formula. Their Lebesgue constant grows only logarithmically with n, so the interpolant is close to the best polynomial approximation. For the Runge function, or any function analytic in a neighborhood of $[-1, 1]$, the error decreases exponentially with n.

A plot of the error $\max_{[-1,1]} |u - p_n|$ against n for the Runge function shows exponential growth for equally spaced nodes and exponential decay for Chebyshev nodes. △

💻 **Implementation exercise 1** (Interpolation, **5 marks**)**.** Write Julia code that computes and plots the interpolating polynomial $p \in \mathcal{P}_3$ through the following points: $(0, 0)$, $(1, 4)$, $(2, 15)$, $(3, 40)$. The plot should display both the interpolation points and the graph of the interpolating polynomial over an appropriate range. Do not use any other library than the ones already imported.

Solution. We solve the Vandermonde system for the coefficients of $p(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3$.

```
using LinearAlgebra
using Plots
x = [0, 1, 2, 3]
y = [0, 4, 15, 40]
V = [xi^j for xi in x, j in 0:3]  # Vandermonde matrix
  = V \ y
p(t) = sum( [j+1] * t^j for j in 0:3)
t = range(-0.5, 3.5, length=200)
plot(t, p.(t), label="Interpolating polynomial")
scatter!(x, y, label="Interpolation points")
```

The interpolating polynomial is $p(x) = \frac{17}{6}x + \frac{7}{6}x^3$. △

⚙ **Question 4** (Numerical integration, **5 marks**)**.** Are the following statements true or false? Justify briefly.

1. The degree of precision of the following quadrature rule is 2:

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx 2u(0)\,.$$

2. The degree of precision of the following rule is equal to 3:

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx u\left(-\frac{1}{3}\right) + u\left(\frac{1}{3}\right).$$

3. For any natural number $N > 0$, there exists a quadrature rule with a degree of precision equal to $2N-1$ of the form

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx \sum_{n=1}^{N} w_n u(x_n).$$

4. Let $x_i^N = i/N$ and consider the following approximation of $\int_0^1 u(x)\,\mathrm{d}x$:

$$\widehat{I}_N = \frac{1}{2N}\Big(u\left(x_0^N\right) + 2u\left(x_1^N\right) + 2u\left(x_2^N\right) + \ldots + 2u\left(x_{N-2}^N\right) + 2u\left(x_{N-1}^N\right) + u\left(x_N^N\right)\Big). \tag{1}$$

   Suppose first that u is the Runge function, given by $u(x) = (1+25x^2)^{-1}$. Then $\widehat{I}_N$ diverges in the limit $N \to \infty$.

5. Let $u(x) = \cos(3x)$ and let $\widehat{I}_N$ be as in (1). Then it holds that

$$\lim_{N\to+\infty}\left(\left|\widehat{I}_N - \int_0^1 u(x)\,\mathrm{d}x\right|\right) = 0.$$

6. (**Bonus.**) Fix $u(x) = 2x-1$ and let $\widehat{I}_N$ be as in (1). Then $\widehat{I}_N = 0$ for all $N \geqslant 2$.

Solution.

1. **False**: the rule is exact for $u(x) = 1$ and $u(x) = x$, but not for $u(x) = x^2$, since $\int_{-1}^1 x^2\,\mathrm{d}x = \frac{2}{3}$ whereas $2 \times 0^2 = 0$. Its degree of precision is 1.
2. **False**: the rule is exact for $u(x) = 1$ (both sides equal 2) and $u(x) = x$ (both sides equal 0), but for $u(x) = x^2$ the rule gives $\frac{2}{9} \neq \frac{2}{3}$. Its degree of precision is 1. The rule with degree of precision 3 is the Gauss–Legendre rule with nodes $\pm 1/\sqrt{3}$.
3. **True**: the Gauss–Legendre rule with N nodes, whose nodes are the roots of the Legendre polynomial of degree N, has degree of precision $2N-1$.
4. **False**: $\widehat{I}_N$ is the composite trapezium rule. Since the Runge function is smooth on $[0,1]$, the error satisfies $\left|\widehat{I}_N - \int_0^1 u(x)\,\mathrm{d}x\right| \leqslant \frac{1}{12N^2}\max_{[0,1]}|u''|$, so $\widehat{I}_N$ converges to the integral. (Runge's phenomenon concerns interpolation by a single polynomial of high degree, not composite rules.)
5. **True**, by the same error bound: the error is at most $\frac{9}{12N^2}$, since $|u''| \leqslant 9$.
6. **True**: the trapezium rule is exact for polynomials of degree 1 on each subinterval, so $\widehat{I}_N = \int_0^1 (2x-1)\,\mathrm{d}x = 0$ for all $N \geqslant 1$. This can also be checked directly, since $u(x_i^N) + u(x_{N-i}^N) = 0$ and the weights are symmetric.

△

Question 5 (Gaussian–Hermite numerical integration, **10 marks**)**.** The Gauss–Hermite quadra-

ture formula with n nodes is an approximation of the form

$$I(u) \coloneqq \int_{-\infty}^{\infty} u(x)\,\mathrm{e}^{-x^2}\,\mathrm{d}x \approx \sum_{i=1}^{n} w_i\,u(x_i) =: \widehat{I}_n(u),$$

which is exact when u is a polynomial of degree $\leq 2n-1$. Note that the nodes are numbered $1,\dots,n$. For this question, we take for granted that, for integers $i \geq 0$, it holds that

$$\int_{-\infty}^{\infty} x^i \mathrm{e}^{-x^2}\,\mathrm{d}x = \begin{cases} 0, & \text{if } i \text{ is odd,} \\ (i-1)!!\,\sqrt{\dfrac{\pi}{2^i}}, & \text{if } i \text{ is even,} \end{cases}$$

where $(i-1)!! \coloneqq 1 \times 3 \times 5 \times \cdots \times (i-1)$. In particular, with all the integrals being over $(-\infty,\infty)$, the following special cases may be useful in your computations:

$$\int \mathrm{e}^{-x^2}\,\mathrm{d}x = \sqrt{\pi}, \quad \int x^2\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}, \quad \int x^4\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \frac{3}{4}\sqrt{\pi}, \quad \int x^6\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \frac{15}{8}\sqrt{\pi}\,.$$

1. (**5 marks**) Find the nodes and weights of the Gauss–Hermite rule with $n = 3$ nodes. By symmetry, we expect nodes of the form $(-z, 0, z)$ and weights (w_1, w_2, w_1), which reduces the number of unknowns to three.

   Solution. By symmetry, the rule is exact for all odd monomials. Requiring exactness for 1, x^2 and x^4 gives

   $$2w_1 + w_2 = \sqrt{\pi}, \qquad 2w_1 z^2 = \frac{\sqrt{\pi}}{2}, \qquad 2w_1 z^4 = \frac{3}{4}\sqrt{\pi}.$$

   Dividing the third equation by the second gives $z^2 = \frac{3}{2}$, and then $w_1 = \frac{\sqrt{\pi}}{4z^2} = \frac{\sqrt{\pi}}{6}$ and $w_2 = \sqrt{\pi} - 2w_1 = \frac{2\sqrt{\pi}}{3}$. The nodes and weights are therefore

   $$(x_1, x_2, x_3) = \left(-\sqrt{\tfrac{3}{2}}, 0, \sqrt{\tfrac{3}{2}}\right), \qquad (w_1, w_2, w_3) = \left(\frac{\sqrt{\pi}}{6}, \frac{2\sqrt{\pi}}{3}, \frac{\sqrt{\pi}}{6}\right).$$

   The rule is exact for all polynomials of degree at most $5 = 2n-1$, but not for x^6: it gives $2w_1 z^6 = \frac{9}{8}\sqrt{\pi} \neq \frac{15}{8}\sqrt{\pi}$. △

2. (**5 marks**) Let $\{H_0, H_1, \dots\}$ denote orthogonal polynomials for the inner product

   $$\langle f, g\rangle := \int_{-\infty}^{\infty} f(x)g(x)\,\mathrm{e}^{-x^2}\,\mathrm{d}x$$

   which, in addition, satisfy the following two conditions:

   - For all $i \in \mathbf{N}$, the polynomial H_i is of degree i.
   - The leading coefficient of H_i, which multiplies x^i, is equal to 1.

   Calculate H_0, H_1, H_2 and H_3. What is the relationship between H_3 and the quadrature rule found in the first item?

Solution. Since the weight e^{-x^2} is even, H_i is even when i is even and odd when i is odd, so all odd-degree (resp. even-degree) inner products vanish automatically. We obtain

- $H_0(x) = 1$;
- $H_1(x) = x$, since $\langle x, 1\rangle = 0$;
- $H_2(x) = x^2 - a$ with $\langle x^2 - a, 1\rangle = \frac{\sqrt{\pi}}{2} - a\sqrt{\pi} = 0$, so $H_2(x) = x^2 - \frac{1}{2}$;
- $H_3(x) = x^3 - bx$ with $\langle x^3 - bx, x\rangle = \frac{3}{4}\sqrt{\pi} - b\frac{\sqrt{\pi}}{2} = 0$, so $H_3(x) = x^3 - \frac{3}{2}x$.

The roots of $H_3(x) = x\big(x^2 - \frac{3}{2}\big)$ are 0 and $\pm\sqrt{3/2}$: they are exactly the nodes of the Gauss–Hermite rule with 3 nodes. This is a general fact: the nodes of the Gauss quadrature rule with n nodes are the roots of the orthogonal polynomial of degree n for the corresponding weight. △

3. (Bonus, **2 marks**) Calculate H_4 and, using this result, deduce the nodes and weights of the Gauss–Hermite quadrature with 4 points.

Solution. By parity, $H_4(x) = x^4 + bx^2 + c$, and orthogonality to 1 and x^2 gives

$$\frac{3}{4} + \frac{b}{2} + c = 0, \qquad \frac{15}{8} + \frac{3b}{4} + \frac{c}{2} = 0.$$

Subtracting the first equation from twice the second gives $3 + b = 0$, so $b = -3$, $c = \frac{3}{4}$ and $H_4(x) = x^4 - 3x^2 + \frac{3}{4}$. Its roots satisfy $x^2 = \frac{3\pm\sqrt{6}}{2}$, so the nodes of the rule are $\pm a$ and $\pm b$ with

$$a = \sqrt{\frac{3-\sqrt{6}}{2}} \approx 0.5246, \qquad b = \sqrt{\frac{3+\sqrt{6}}{2}} \approx 1.6507.$$

By symmetry, the weights are of the form (w_b, w_a, w_a, w_b), and the rule is exact for odd polynomials. Exactness for 1 and x^2 gives $w_a + w_b = \frac{\sqrt{\pi}}{2}$ and $w_a a^2 + w_b b^2 = \frac{\sqrt{\pi}}{4}$, whose solution is

$$w_a = \frac{3+\sqrt{6}}{12}\sqrt{\pi} \approx 0.8049, \qquad w_b = \frac{3-\sqrt{6}}{12}\sqrt{\pi} \approx 0.0813.$$

Since the nodes are the roots of H_4, the rule is then exact for all polynomials of degree at most 7. △

💻 **Implementation exercise 2** (Numerical integration, **10 marks**)**.** The midpoint quadrature rule reads

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx 2u(0)\,.$$

- (**3 marks**) Write a function `midpoint(u, a, b)` that returns, using this quadrature rule, an approximation of the integral

$$\int_a^b u(x)\,\mathrm{d}x\,. \tag{2}$$

- (**4 marks**) Write a function `composite_midpoint(u, a, b, N)` that returns an approximation of the integral (2), this time using a composite version of the midpoint rule. More precisely, the approximation should be obtained by partitioning the integration interval $[a, b]$ into N cells, and applying the midpoint rule within each cell.

- (**3 marks**) Take $u(x) = \cos(x)$, $a = -1$ and $b = 1$. In this setting, plot the evolution of the error for N varying from 1 to 1000.

Solution. By the change of variable $x = \frac{a+b}{2} + \frac{b-a}{2}t$, the midpoint rule on $[a, b]$ reads $\int_a^b u(x)\,\mathrm{d}x \approx (b-a)\,u\big(\frac{a+b}{2}\big)$.

```
function midpoint(u, a, b)
    return (b - a) * u((a + b) / 2)
end

function composite_midpoint(u, a, b, N)
    h = (b - a) / N
    return sum(midpoint(u, a + (i-1)*h, a + i*h) for i in 1:N)
end

using Plots
u(x) = cos(x)
I = 2sin(1)  # Exact value of the integral
Ns = 1:1000
errors = [abs(composite_midpoint(u, -1, 1, N) - I) for N in Ns]
plot(Ns, errors, xscale=:log10, yscale=:log10,
     xlabel="N", ylabel="Error", label="Composite midpoint rule")
plot!(Ns, Ns.^(-2), linestyle=:dash, label="1/N²")
```

In logarithmic scale, the error decreases along a line of slope -2: the composite midpoint rule converges at rate $\mathcal{O}(N^{-2})$. More precisely, the error is approximately $\frac{h^2}{24}\big|u'(b) - u'(a)\big| = \frac{\sin(1)}{3N^2}$ with $h = 2/N$. Note that the code template in the exam statement omits the argument `N` of `composite_midpoint`. △